

Machine Problem #1: Managing a Project with Git

Set up your Git username and email (if not already configured)

```
Leila Montemayor@ARCALIENWARE14 MINGW64 ~/ACADS/Montemayor_local_project
$ git config --global user.name "leila-montemayor"

Leila Montemayor@ARCALIENWARE14 MINGW64 ~/ACADS/Montemayor_local_project
$ git config --global user.email "lmontemayor.k12045986@umak.edu.ph"

Leila Montemayor@ARCALIENWARE14 MINGW64 ~/ACADS/Montemayor_local_project
$ git config --global --list
user.email=lmontemayor.k12045986@umak.edu.ph
user.name=leila-montemayor
```

```
$ git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=openssl
http.sslcainfo=C:/Program Files/Git/mingw64/etc/ssl/certs/ca-bundle.crt
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.rebase=false
credential.helper=manager
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=master
user.email=lmontemayor.k12045986@umak.edu.ph
user.name=leila-montemayor
core.repositoryformatversion=0
core.filemode=false
core.bare=false
core.logallrefupdates=true
core.symlinks=false
core.ignorecase=true
```

Create a directory called <YourLastname>_local_project, navigate into it, and initialize it as a Git repository

```
Leila Montemayor@ARCALIENWARE14 MINGW64 ~/ACADS/Montemayor_local_project
$ git init
Initialized empty Git repository in C:/Users/Leila Montemayor/ACADS/Montemayor_lo
cal_project/.git/
```

Tasks:

1. Create a new branch:

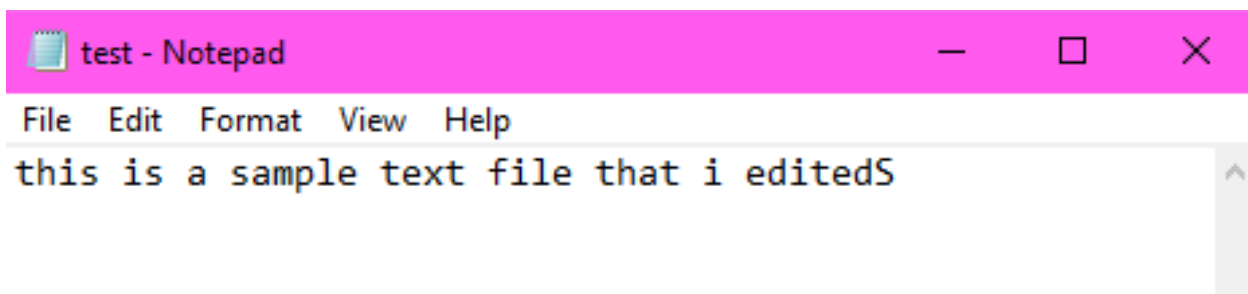
- a. Create a new branch named "feature_x" to isolate your experimental changes.

```
Leila Montemayor@ARCALIENWARE14 MINGW64 ~/ACADS/Montemayor_local_project (master)
$ git branch feature_x

Leila Montemayor@ARCALIENWARE14 MINGW64 ~/ACADS/Montemayor_local_project (master)
$ git checkout feature_x
Switched to branch 'feature_x'

Leila Montemayor@ARCALIENWARE14 MINGW64 ~/ACADS/Montemayor_local_project (feature_x)
$
```

2. Stage changes:
 - a. Make modifications to your project files.



- b. Stage the changes using the git add command.

```
Leila Montemayor@ARCALIENWARE14 MINGW64 ~/ACADS/Montemayor_local_project (feature_x)
$ git add .

Leila Montemayor@ARCALIENWARE14 MINGW64 ~/ACADS/Montemayor_local_project (feature_x)
$ git status
On branch feature_x
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   test.txt
```

3. Commit changes:
 - a. Commit the staged changes to your local repository using the git commit command. Provide a clear and concise commit message.

```
Leila Montemayor@ARCALIENWARE14 MINGW64 ~/ACADS/Montemayor_local_project (feature_x)
$ git commit -m "edited content of text.txt file"
[feature_x 7f4bdbd] edited content of text.txt file
1 file changed, 1 insertion(+), 1 deletion(-)
```

4. Switch back to the main branch:
 - a. Use the git checkout command to switch back to the main branch.

```
Leila Montemayor@ARCALIENWARE14 MINGW64 ~/ACADS/Montemayor_local_project (feature_x)
$ git checkout master
Switched to branch 'master'

Leila Montemayor@ARCALIENWARE14 MINGW64 ~/ACADS/Montemayor_local_project (master)
$
```

5. Merge changes from the feature branch:
 - a. Use the git merge command to merge the changes from the "feature_x" branch into the main branch. Resolve any merge conflicts that may arise.

```
Leila Montemayor@ARCALIENWARE14 MINGW64 ~/ACADS/Montemayor_local_project (master)
$ git merge feature_x
Updating 9663374..7f4bdbd
Fast-forward
 test.txt | 2 +-
 1 file changed, 1 insertion(+), 1 deletion(-)
```

6. Viewing the Commit History:
 - a. Use Git commands to view the commit history and ensure that all commits are in place.

```
Leila Montemayor@ARCALIENWARE14 MINGW64 ~/ACADS/Montemayor_local_project (master)
$ git log
commit 7f4bdbd543c394d9d2dc6a2da12b95774a01a327 (HEAD -> master, feature_x)
Author: leila-montemayor <lmontemayor.k12045986@umak.edu.ph>
Date:   Fri Sep 19 10:35:25 2025 +0800

    edited content of text.txt file

commit 9663374ebf5a67526b49570e1bfebf5493ed9b4f
Author: leila-montemayor <lmontemayor.k12045986@umak.edu.ph>
Date:   Fri Sep 19 10:29:26 2025 +0800

    initial commit

Leila Montemayor@ARCALIENWARE14 MINGW64 ~/ACADS/Montemayor_local_project (master)
$ |
```

7. Tagging Versions (Optional):
 - a. After completing the merge, use the git tag command to create a tag named v1.0 to mark the first version of your project.

```
Leila Montemayor@ARCALIENWARE14 MINGW64 ~/ACADS/Montemayor_local_project (master)
$ git tag v1.0

Leila Montemayor@ARCALIENWARE14 MINGW64 ~/ACADS/Montemayor_local_project (master)
$ git log
commit 7f4bdbc543c394d9d2dc6a2da12b95774a01a327 (HEAD -> master, tag: v1.0, feature_x)
Author: leila-montemayor <lmontemayor.k12045986@umak.edu.ph>
Date:   Fri Sep 19 10:35:25 2025 +0800

    edited content of text.txt file

commit 9663374ebf5a67526b49570e1bfebf5493ed9b4f
Author: leila-montemayor <lmontemayor.k12045986@umak.edu.ph>
Date:   Fri Sep 19 10:29:26 2025 +0800

    initial commit
```